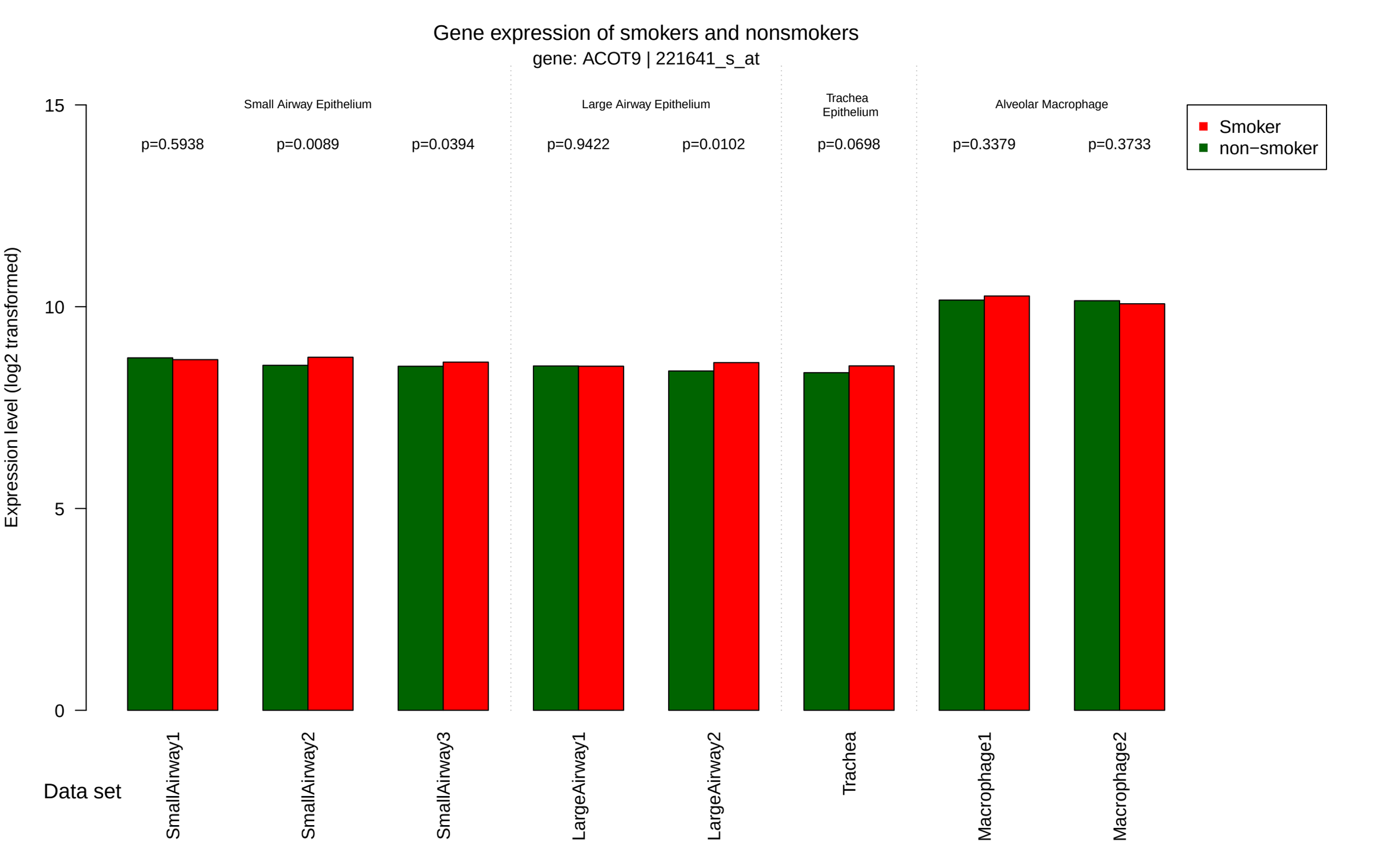
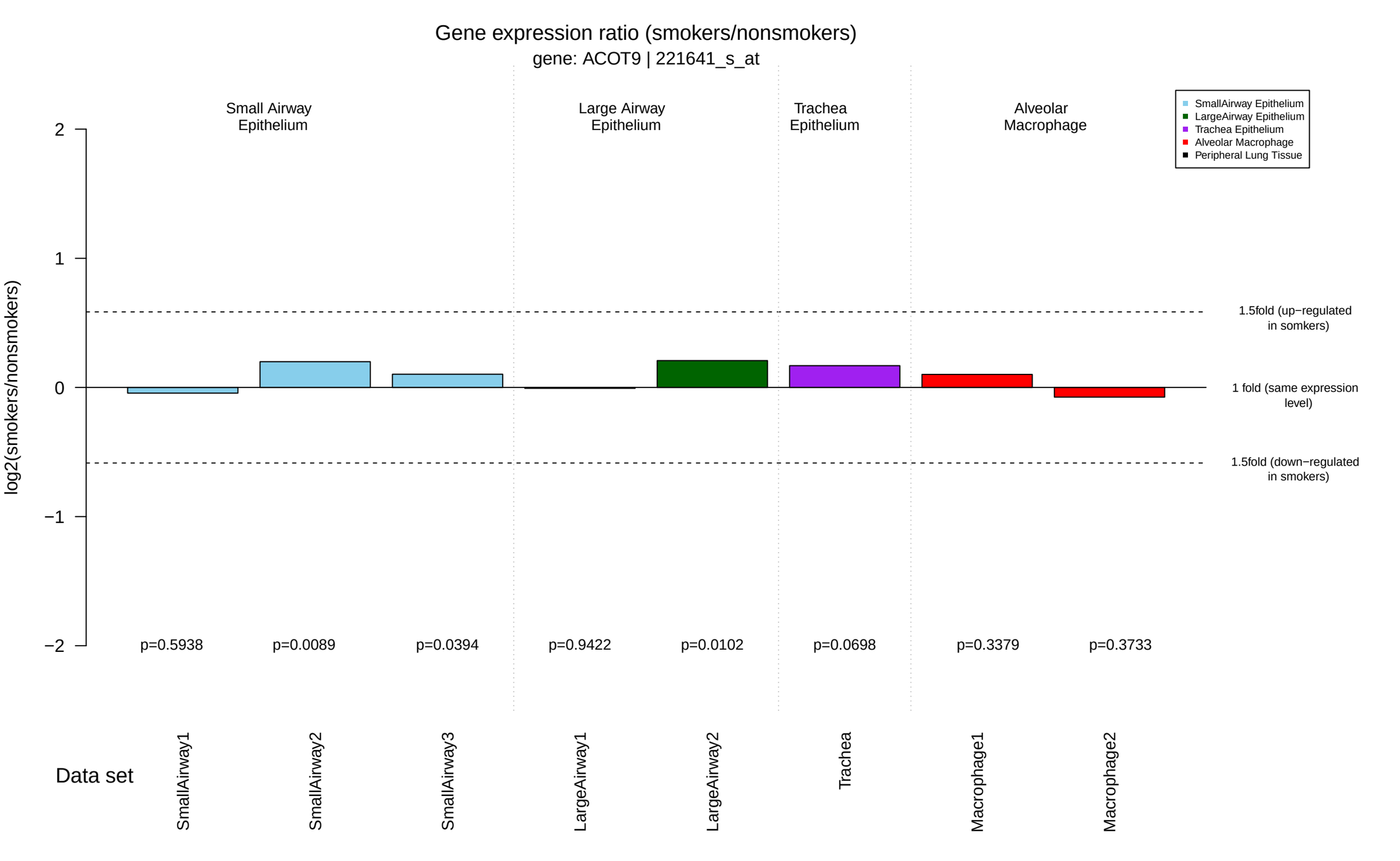
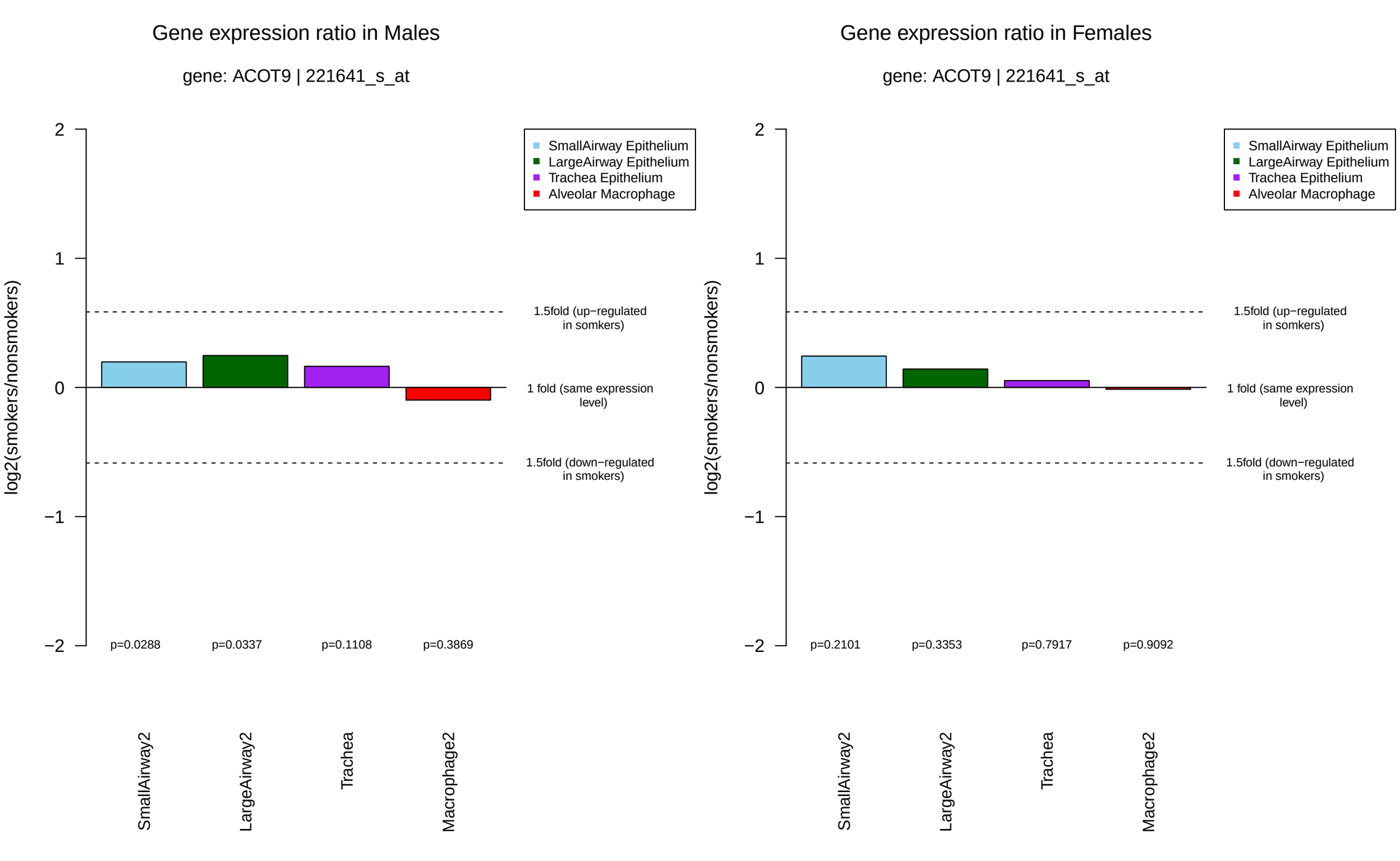
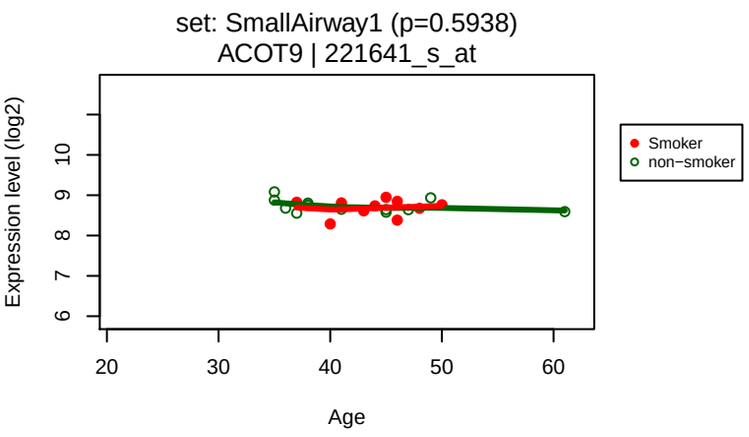
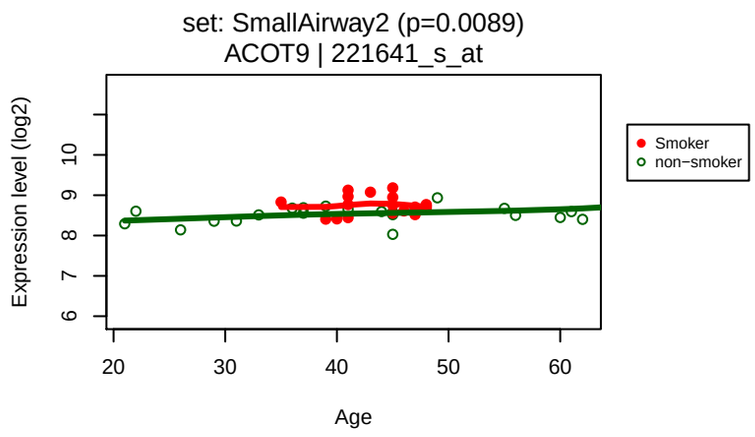
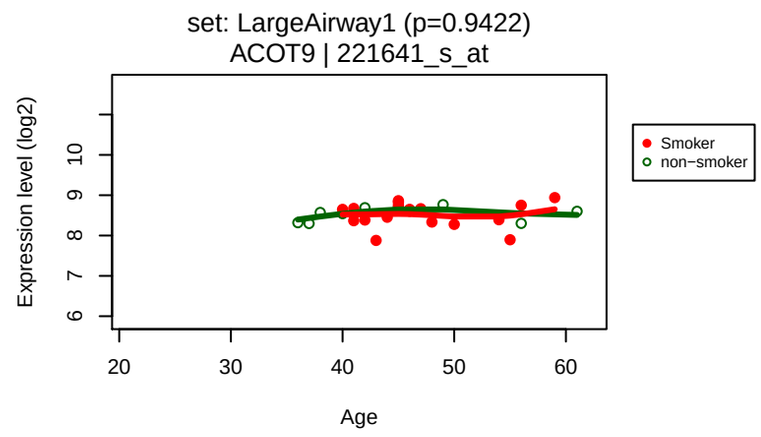

Gene expression of smokers and nonsmokers
gene: ACOT9 | 221641_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5938
p=0.0089
p=0.0394
p=0.9422
p=0.0102
p=0.0698
p=0.3379
p=0.3733
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

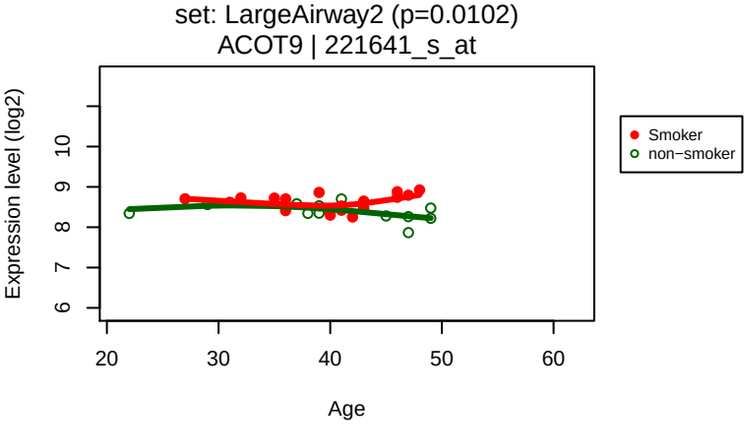

Gene expression ratio (smokers/nonsmokers)
gene: ACOT9 | 221641_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5938
p=0.0089
p=0.0394
p=0.9422
p=0.0102
p=0.0698
p=0.3379
p=0.3733
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

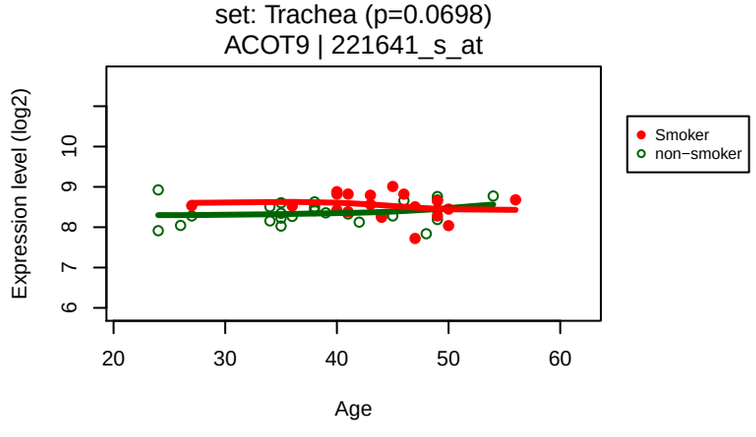

Gene expression ratio in Males
gene: ACOT9 | 221641_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0288
p=0.0337
p=0.1108
p=0.3869
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACOT9 | 221641_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2101
p=0.3353
p=0.7917
p=0.9092
SmallAirway2
LargeAirway2
Trachea
Macrophage2

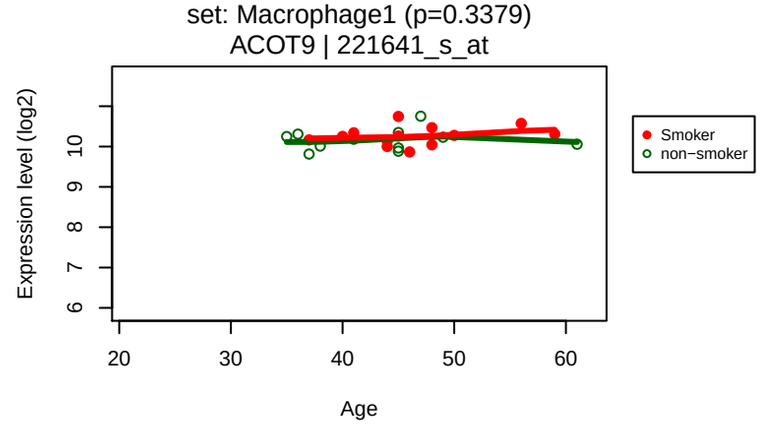

set: SmallAirway1 (p=0.5938)
ACOT9 | 221641_s_at
Expression level (log2)
Age
Smoker
non−smoker

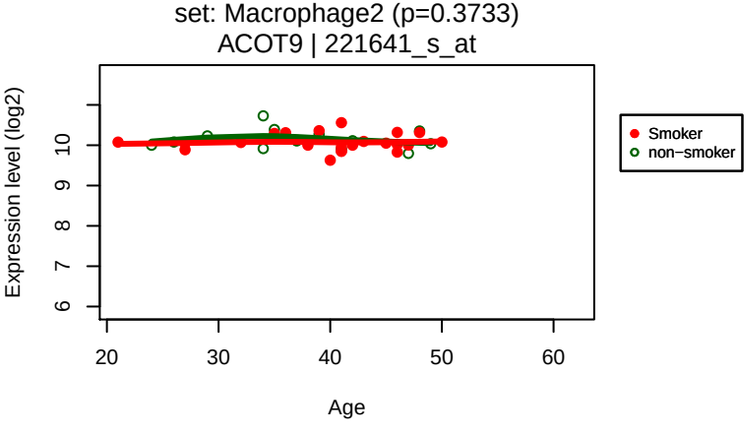

set: SmallAirway2 (p=0.0089)
ACOT9 | 221641_s_at
Expression level (log2)
Age
Smoker
non−smoker

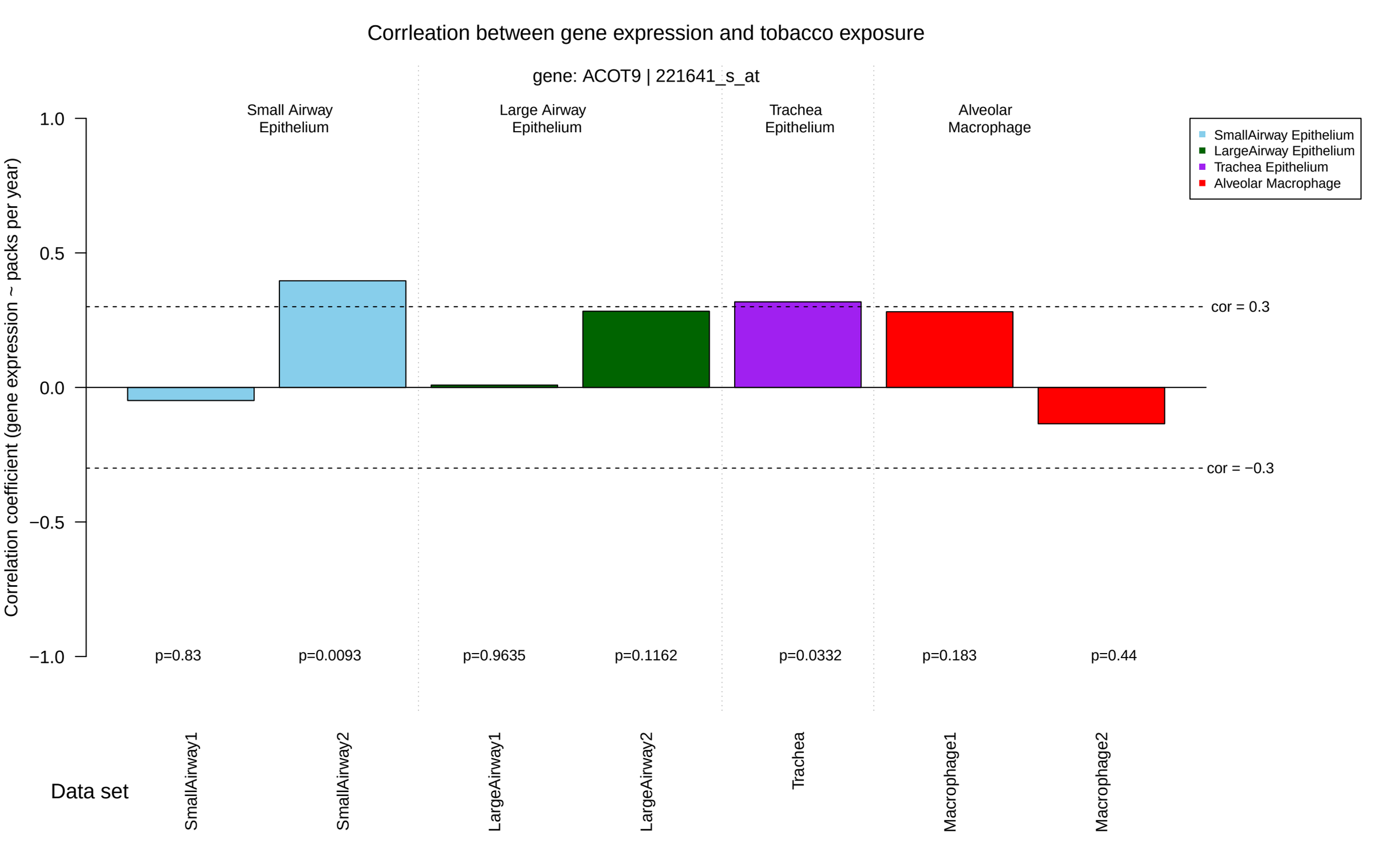

set: LargeAirway1 (p=0.9422)
ACOT9 | 221641_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0102)
ACOT9 | 221641_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0698)
ACOT9 | 221641_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3379)
ACOT9 | 221641_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3733)
ACOT9 | 221641_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACOT9 | 221641_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.83
p=0.0093
p=0.9635
p=0.1162
p=0.0332
p=0.183
p=0.44
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2